SMARTHIKA AGRO

App Userflow & State Reference

GRC_SMARTHIKA_SPI_VI_V3 • Schema v2.1 • April 2026

Elcinotech Solutions Pvt. Ltd. • Internal Document

1. Document Overview

This document defines the complete userflow for the SMARTHIKA mobile application. It covers every screen state, every LED sequence payload, every LCD mirror state, and every edge case including OTA firmware upgrade. The app is a live remote mirror of the physical GRC controller — the same 10 LEDs, the same 16×2 LCD display, and the same 4 buttons (START, STOP, MODE, SET) are replicated exactly in the app UI.

The app does not drive the device directly. All state changes flow through MQTT over AWS IoT Core. The app reads LED states from an MQTT LED sequence payload and mirrors the LCD text exactly as published by the firmware.

1.1 System Architecture — App Layer

Layer	Responsibility	MQTT Topics Used
Firmware	Measure, protect, publish telemetry. Control relay. Manage LCD & LEDs on device.	gw/{id}/telemetry, status, health, error, response
MQTT Broker	AWS IoT Core ap-south-1. Routes all messages. Enforces QoS.	—
Cloud Lambda	Borewell science, duty cycle, yield test decisions. Publishes commands.	gw/{id}/cmd, leds
App	Mirror LCD & LEDs. Send button tap commands. Show dashboard data.	gw/{id}/leds, telemetry, status, health, error

1.2 The 10 LEDs — Identity Reference

Index	LED Name	Color	What It Represents	MQTT Source
0	LINE 1 (Phase R)	Red	Phase R / Line 1 voltage present and healthy	Computed from el.v[0]
1	LINE 2 (Phase Y)	Amber	Phase Y / Line 2 voltage present and healthy	Computed from el.v[1]
2	LINE 3 (Phase B)	Blue	Phase B / Line 3 voltage present and healthy	Computed from el.v[2]
3	MOTOR ON	Green	Motor contactor energised (running or starting)	From m.st in telemetry
4	MOTOR TRIP	Red	Motor fault / protection trip active	From gw/{id}/error or m.st=fault
5	SMARTHIKA MODE	Cyan	Device is in Auto / Smart mode	From mode in status topic
6	MANUAL MODE	Orange	Device is in Manual mode	From mode in status topic
7	NET 1	Green	Primary network (GSM) connected	From net.connected in health
8	NET 2	Green	Secondary network (LoRa) present	From health / GSM:OK LORA:OK
9	NET 3	Green	Reserved / tertiary network (future)	Reserved

1.3 LED Sequence MQTT Topic

The cloud publishes LED states on a dedicated topic after every significant state change. The app subscribes to this topic and updates all 10 LEDs atomically.

```
Topic:  gw/{gw_id}/leds
 QoS 1  •  Published by Cloud Lambda on every motor/mode/network state change
  Payload example:
 {
   "leds": [1, 1, 1, 0, 0, 1, 0, 1, 0, 0],
   "ts": "20260413T160000Z"
 }
  Array index maps directly to LED index 0-9 above. 1 = ON, 0 = OFF.
```

> *The app must never infer LED states from other topics. It only renders what the leds payload says. This keeps the app a pure mirror and prevents state drift.*

2. Boot Sequence & OTA Firmware Upgrade

2.1 Device Boot — What the App Shows

When the device powers on it runs through 4 LCD screens before entering normal operation. The app mirrors these exactly. The app receives the LCD text via the existing telemetry and health topics — the firmware publishes its boot state in the health topic on first connect.

Boot Screen A — Splash (~3 seconds)

```
GRC SMARTHIKA
SMART AGRO V3.0
```

LINE 1	LINE 2	LINE 3	MTR ON	TRIP	SMART	MANUAL	NET 1	NET 2	NET 3

> *All LEDs OFF during splash. App shows connecting spinner over the LED panel.*

Boot Screen B — Hardware Diagnostics (1 second)

```
SYSTEM DIAG...
LCD:OK LED:OK
```

LINE 1	LINE 2	LINE 3	MTR ON	TRIP	SMART	MANUAL	NET 1	NET 2	NET 3

> *All LEDs still OFF. App mirrors text only.*

Boot Screen C — Network Diagnostics (1 second)

```
GSM:OK LORA:OK
PRI:OK SEC:OK
```

LINE 1	LINE 2	LINE 3	MTR ON	TRIP	SMART	MANUAL	NET 1	NET 2	NET 3

> *All LEDs OFF. If GSM or LoRa shows FAIL, the app shows a warning banner below the LCD panel.*

Boot Screen D — System Ready (0.8 seconds)

```
SMARTHIKA AGRO
System OK...
```

> *NET 1 LED turns ON once firmware confirms GSM hardware present. This is the first LED state published on gw/{id}/leds.*

2.2 GSM Wait Screen (up to 120 seconds)

After boot the device waits for GSM to connect. The LCD Row 0 is static. Row 1 changes dynamically. The app mirrors all three states:

GSM State	LCD Row 0	LCD Row 1	LED Sequence
Connecting (default)	SMARTHIKA AGRO	Connecting GSM..	leds: [0,0,0,0,0,0,0,0,0,0] — all off
Connected	SMARTHIKA AGRO	System OK	leds: [1,1,1,0,0,1,0,1,1,0] — phases+smart+net1+net2
Error / Timeout	SMARTHIKA AGRO	Manual Mode	leds: [1,1,1,0,0,0,1,0,0,0] — phases+manual only

> *If the user presses STOP during GSM wait, the device skips to Manual Mode. The app must show a toast: "Device entered Manual Mode — GSM bypassed."*

2.3 OTA Firmware Upgrade — 4:00 AM Daily Check

Every morning at 4:00 AM the device checks for a firmware update. The check runs before normal operation resumes. This is the only time the device may go silent or show non-operational LCD states.

2.3.1 OTA Flow — What Happens

1. At 4:00 AM the device publishes a health payload with sys.fw = current version.
2. Cloud Lambda checks firmware registry. If version matches latest, no action — device proceeds normally.
3. If upgrade available, cloud publishes OTA_START command on gw/{id}/cmd.
4. Device enters OTA mode. LCD changes to upgrade screen. Telemetry stops.
5. After successful flash, device reboots and runs full boot sequence (Section 2.1).
6. If OTA fails, device reboots into existing firmware and logs error. Normal operation resumes.

2.3.2 LCD States During OTA

```
FW UPGRADE...
DO NOT POWER OFF
```

During active upgrade:

- LCD Row 0: FW UPGRADE...
- LCD Row 1: DO NOT POWER OFF
- All LEDs OFF — cloud publishes leds: [0,0,0,0,0,0,0,0,0,0] before OTA begins

```
FW v1.2.0 OK
Rebooting...
```

On successful upgrade completion before reboot:

- LCD Row 0: FW v{new_version} OK
- LCD Row 1: Rebooting...

```
FW UPGRADE FAIL
Running v1.1.0
```

On upgrade failure:

- LCD Row 0: FW UPGRADE FAIL
- LCD Row 1: Running v{current_version}

2.3.3 App Behaviour During OTA

Condition	App Display	Duration
4:00–4:05 AM, no OTA needed	Normal operation — no change	Instant
OTA command published by cloud	LCD panel shows FW UPGRADE screen. All LEDs go dark. Yellow banner: "Firmware upgrade in progress — do not send commands."	Until reboot
Device silent 4:00–4:30 AM + health showed version mismatch	Same OTA banner — treat as upgrade in progress, not offline	Up to 30 min

Condition	App Display	Duration
Device silent 4:00–4:30 AM + no prior version mismatch	OTA banner for first 20 min, then "Device Offline" warning	After 20 min
Device silent beyond 4:30 AM	"Device Offline" alert regardless of OTA context	Persistent until reconnect
Device reconnects post-OTA	OTA banner clears. Boot sequence mirrors (Section 2.1). Normal operation resumes.	Instant on reconnect

> *During OTA the START, STOP, MODE, and SET buttons in the app must be disabled. Tapping any button shows toast: "Commands disabled — firmware upgrade in progress."*

3. Idle State — Motor Stopped

When the motor is idle the LCD auto-rotates through 4 slots every 4 seconds. The app mirrors this rotation exactly — same slot, same timing, driven by the LCD text published in the telemetry topic. The app does not run its own timer; it renders whatever text the firmware is currently publishing.

3.1 LED Sequence — Idle, Auto Mode, All Phases Present

leds: [1, 1, 1, 0, 0, 1, 0, 1, 1, 0]

- LINE 1, 2, 3 ON — all three phases healthy
- MOTOR ON OFF — motor not running
- MOTOR TRIP OFF — no fault
- SMARTHIKA MODE ON — auto mode active
- MANUAL MODE OFF
- NET 1 ON — GSM connected
- NET 2 ON — LoRa present
- NET 3 OFF — reserved

3.2 LCD Auto-Rotation Slots — Idle

Slot A — Branding (every 4s)

```
SMARTHIKA AGRO
System OK
```

Slot B — Network Status

```
GSM: Connected
PRI:ONLINE SEC:-
```

> *If GSM searching: Row 0 = "GSM: Searching.." — NET 1 LED will be OFF in leds payload.*

Slot C — Uptime and Mode

```
Uptime: 02:45
Mode: AUTO
```

Slot D — Voltage and Frequency

```
V:231V F:50Hz
```

```
SYSTEM STANDBY
```

3.3 Phase Loss Scenarios — Idle

Condition	LED Change	LCD Row 0 / Row 1 (Slot D)
Phase R missing	LINE 1 OFF, others unchanged	V:---V F:50Hz / PHASE R MISSING
Phase Y missing	LINE 2 OFF, others unchanged	V:231V F:50Hz / PHASE Y MISSING
Phase B missing	LINE 3 OFF, others unchanged	V:231V F:50Hz / PHASE B MISSING
All phases lost	LINE 1,2,3 OFF	V:---V F:--- / NO SUPPLY

> *Phase status is derived by Cloud Lambda from el.v[0..2] in telemetry. Lambda publishes updated leds payload. App renders accordingly.*

3.4 Idle — Manual Mode

leds: [1, 1, 1, 0, 0, 0, 1, 1, 1, 0]

- SMARTHIKA MODE OFF, MANUAL MODE ON
- All other states same as Auto idle

> *In Manual Mode the SET button in the app becomes active. A label appears below the LCD panel: "SET: Short press to navigate pages."*

4. Running State — Motor ON

When the motor starts, the LCD switches to the running home rotation (3 slots, every 3 seconds). The LED sequence updates immediately on motor start.

4.1 Starting Sequence — Star Phase

```
Motor Starting..
Running (Star)
```

leds: [1, 1, 1, 1, 0, 1, 0, 1, 1, 0]

- MOTOR ON turns ON immediately when contactor energises
- MOTOR TRIP remains OFF

> *App shows a pulsing animation on the MOTOR ON LED during star phase to indicate transition in progress.*

4.2 Transitioning — Delta Phase

```
Motor Starting..
Running (Delta)
```

leds: [1, 1, 1, 1, 0, 1, 0, 1, 1, 0] — same as star phase

> *LED sequence does not change between star and delta. Only the LCD text changes. The app must not add any additional visual indicator beyond what the LCD mirror shows.*

4.3 Running — Full Speed (Star-Delta complete)

leds: [1, 1, 1, 1, 0, 1, 0, 1, 1, 0]

Running Slot A — Motor State and Runtime

```
Motor On
Runtime 04:32
```

Running Slot B — Power Monitoring

```
V:230 I: 4.2A
PHASE MONITORING
```

Running Slot C — Session Clock

```
RUNNING TIME:
HH:MM:SS
```

4.4 Single Contactor Mode — Running

```
Motor Running
Runtime 04:32
```

In Single Contactor mode Slot A Row 0 reads "Motor Running" instead of "Motor On". LED sequence is identical to star-delta running.

4.5 Motor Stop

When motor stops the device publishes a status event and new leds payload. The LCD transitions back to idle rotation immediately.

leds: [1, 1, 1, 0, 0, 1, 0, 1, 1, 0] — MOTOR ON turns OFF

> *Cloud starts recovery logging timer immediately on MOTOR ON → OFF transition. App may show a passive indicator "Recovery logging active" in the dashboard section below the LCD panel.*

5. Fault State — System Fault

A fault immediately overrides all LCD screens. The fault screen is system-wide and cannot be dismissed by normal navigation. The app mirrors this override state.

5.1 Fault LED Sequence

leds: [1, 1, 1, 0, 1, 1, 0, 1, 1, 0]

- MOTOR ON turns OFF — relay already latched OFF by firmware
- MOTOR TRIP turns ON — fault active
- Other LEDs reflect current phase and network state

> *For phase fault conditions some phase LEDs may also be OFF — the full state is always carried in the leds payload.*

5.2 Fault LCD

```
SYSTEM FAULT!
Undervoltage
```

Row 1 carries the fault code string from FaultManager. Common values:

Fault Code	LCD Row 1	Phase LEDs Affected	Severity
OVERVOLTAGE_A	Overvoltage R	LINE 1 blinks (cloud sets leds rapidly)	Critical
UNDERVOLTAGE_A/B/C	Undervoltage	Affected phase LED OFF	Critical
OVERCURRENT	Overcurrent	All phase LEDs ON	Critical
ALARM_DRY_RUN	Dry Run	All phase LEDs ON, TRIP ON	Critical
PHASE_LOSS_A/B/C	Phase Loss R/Y/B	Affected phase LED OFF	Critical
PHASE_SEQUENCE	Phase Seq Fault	All phase LEDs ON	Warning

5.3 Fault — App Behaviour

- Red banner across top of app: "SYSTEM FAULT — [fault description]"
- START button disabled and greyed out
- STOP button active (always allowed)
- MODE button disabled
- SET button active — short press clears fault (mirrors physical device)
- Buzzer notification sent to phone if app has notification permission

> *Fault clear via SET press does NOT restart the motor. It only clears the fault latch. The motor must be manually started again after fault clearance.*

5.4 Fault Clear Sequence

7. User presses SET in app.
8. App publishes PUMP_STOP equivalent — not needed as relay already OFF.
9. App sends fault acknowledgment to cloud.
10. Cloud clears fault state. Publishes new leds payload with MOTOR TRIP OFF.
11. LCD returns to idle home rotation.
12. App removes red fault banner. All buttons re-enable.

6. Button Behaviour & MQTT Commands

6.1 Button State Matrix

Button	Auto Mode	Manual Mode	Fault State	OTA in Progress
START	Active — sends PUMP_START	Active — sends PUMP_START	Disabled	Disabled
STOP	Active — sends PUMP_STOP	Active — sends PUMP_STOP	Active	Disabled
MODE	Active — toggles mode	Active — toggles mode	Disabled	Disabled
SET (short)	Locked — error beep + toast	Active — cycles LCD pages	Active — clears fault	Disabled
SET (long 3s)	Locked	Active on Settings page — toggles motor type	Disabled	Disabled

6.2 MQTT Commands Sent by App Buttons

Button	MQTT Command Published	Expected Response
START	gw/{id}/cmd: {"cmd":"PUMP_START","target":"{id}","ts":"..."}	gw/{id}/response: {"result":"ok"} within 15 sec
STOP	gw/{id}/cmd: {"cmd":"PUMP_STOP","target":"{id}","ts":"..."}	gw/{id}/response: {"result":"ok"} within 15 sec
MODE (to Auto)	gw/{id}/cmd: {"cmd":"MODE_CHANGE","params":{"mode":"auto"},"ts":"..."}	leds payload + status event
MODE (to Manual)	gw/{id}/cmd: {"cmd":"MODE_CHANGE","params":{"mode":"manual"},"ts":"..."}	leds payload + status event
SET short (page cycle)	No MQTT — LCD mirrors firmware page change	New LCD text in telemetry
SET long (motor type)	No MQTT in v2.1 — local to device only	New LCD Row 0 in settings page

> *START is always pre-validated by cloud before reaching the device. The app sends the command; the cloud intercepts, validates voltage, SDL level, and connectivity, then forwards PUMP_START to the device only if all checks pass. If cloud rejects, the app receives a NACK and shows the reason.*

6.3 SET Button — Page Cycle in Manual Mode

In Manual mode, short press of SET cycles through pages in this order. The app shows which page is active by highlighting the page name below the LCD panel.

Page #	Page Name	LCD Row 0	LCD Row 1	App Extra
1	STATUS (home)	(auto-rotation)	(auto-rotation)	No highlight
2	NETWORK	Link: GSM/GPRS	NET: CONNECTED	Show NET 1/2 LED status
3	ENERGY	V:230 I:4.2A	PHASE MONITORING	Show power bar
4	SYSTEM INFO	Uptime: HH:MM	Mode: AUTO	Show firmware version
5	MOTOR RUNTIME	Motor Runtime	Total: HH:MM	Show cumulative hours
6	PHASE DETAILS	LINE V: 398V	3-PHASE MONITOR	Show all 3 phases
7	SETTINGS	Motor: Star-Delta	Hold SET 3-4s	Long-press CTA visible

7. Yield Test — LCD & LED States

Yield tests are cloud-initiated only. The farmer never manually triggers a yield test from the app button — it is always scheduled by the cloud (Section 5.1 of the Cloud Intelligence Module). The app reflects the yield test state through LCD and LED changes.

7.1 Yield Test LED Sequence

leds: [1, 1, 1, 1, 0, 1, 0, 1, 1, 0] — same as normal running state

- No unique LED combination for yield test — motor is simply running
- The app distinguishes yield test from normal run via the active_test field in a new status payload (see below)

7.2 Yield Test LCD Screens

Yield Test Phase	LCD Row 0	LCD Row 1	App Banner
Pre-check (cloud verifying SDL)	SMARTHIKA AGRO	System OK (idle rotation)	Yellow: Yield test pre-check
Step 1 running (60 min)	YIELD TEST S1	DDL: 15.5m T:12:45	Blue: Yield Test Step 1/4
Step 2 running	YIELD TEST S2	DDL: 17.2m T:12:45	Blue: Yield Test Step 2/4
Step 3 running	YIELD TEST S3	DDL: 19.1m T:12:45	Blue: Yield Test Step 3/4
Step 4 running	YIELD TEST S4	DDL: 21.3m T:12:45	Blue: Yield Test Step 4/4
Recovery logging (post-test)	RECOVERY LOG	WL:18.5m T:00:30	Green: Recovery logging
Analysis complete	YIELD TEST OK	Sc:3.2 m3/hr/m	Green: Results ready
Test aborted (dry run)	YIELD TEST STOP	DRY RUN ABORT	Red: Test aborted — dry run

7.3 Yield Test — App Behaviour

- During yield test, START and STOP buttons are replaced by a single "Abort Test" button in the app
- "Abort Test" publishes YIELD_TEST_STOP to gw/{id}/cmd
- MODE button is disabled during yield test
- SET navigation is available in Manual mode as normal
- A progress card appears in the dashboard area below the LCD panel showing: current step, elapsed time, current DDL, and drawdown

- On test completion, a "View Results" button appears linking to the yield test report screen

> *The app must not allow the user to navigate away from the home screen during yield test without a confirmation dialog: "Yield test in progress. Leave anyway?" Leaving does not abort the test — it continues on device.*

8. Network & Connectivity States

8.1 GSM Loss During Operation

Condition	LED Change	LCD (Slot B)	App Action
GSM drops while idle	NET 1 OFF	GSM: Searching.. / PRI:-- SEC:--	Amber banner: GSM reconnecting
GSM drops while running	NET 1 OFF	Slot B changes to searching text	Amber banner — do not stop motor
GSM reconnects	NET 1 ON	GSM: Connected / PRI:ONLINE SEC:--	Banner clears
GSM lost > 5 minutes	NET 1 OFF	GSM: Searching..	Orange banner: Check SIM / signal

> *The motor continues running during GSM loss. The firmware operates independently. The app loses real-time data but must not send stop commands due to connectivity loss alone.*

8.2 Device Offline — App Cannot Reach Device

Duration Silent	App Display
0–20 sec	Last known state shown. Subtle "Updating..." spinner on LCD panel.
20 sec – 2 min	LCD panel greys out. Yellow banner: "Device not responding". Buttons remain enabled but warn on tap.
2 min – 10 min	Red banner: "Device offline". All command buttons disabled except STOP (always try). LED panel shows last known state with 50% opacity.
> 10 min	Full offline screen overlaid on LCD/LED panel. "Last seen: HH:MM". Support contact visible.
Reconnects	All panels restore. Fresh leds payload renders. Banner clears automatically.

9. Complete LED Sequence Reference

This table is the single source of truth for every LED state the app must render. Index: 0=LINE1, 1=LINE2, 2=LINE3, 3=MTR ON, 4=TRIP, 5=SMART, 6=MANUAL, 7=NET1, 8=NET2, 9=NET3

State	leds[] array	Notes
Boot / OTA in progress	[0,0,0,0,0,0,0,0,0,0]	All off — published at boot start and before OTA
GSM connecting	[0,0,0,0,0,0,0,0,0,0]	Phases not yet confirmed
Idle, Auto, 3-phase, GSM+LoRa	[1,1,1,0,0,1,0,1,1,0]	Normal standby state
Idle, Auto, 3-phase, GSM only	[1,1,1,0,0,1,0,1,0,0]	LoRa absent or fault
Idle, Manual, 3-phase, GSM+LoRa	[1,1,1,0,0,0,1,1,1,0]	Manual standby
Idle, Auto, Phase R missing	[0,1,1,0,0,1,0,1,1,0]	LINE 1 off
Idle, Auto, Phase Y missing	[1,0,1,0,0,1,0,1,1,0]	LINE 2 off
Idle, Auto, Phase B missing	[1,1,0,0,0,1,0,1,1,0]	LINE 3 off
Running, Auto, Star or Delta	[1,1,1,1,0,1,0,1,1,0]	MOTOR ON lit
Running, Manual	[1,1,1,1,0,0,1,1,1,0]	MANUAL MODE lit, MOTOR ON lit
Fault, Auto mode	[1,1,1,0,1,1,0,1,1,0]	TRIP on, MOTOR ON off
Fault, Manual mode	[1,1,1,0,1,0,1,1,1,0]	TRIP on, MANUAL lit
Fault, Phase R loss	[0,1,1,0,1,1,0,1,1,0]	LINE 1 off + TRIP on
GSM lost, Idle, Auto	[1,1,1,0,0,1,0,0,1,0]	NET 1 off
GSM lost, Running, Auto	[1,1,1,1,0,1,0,0,1,0]	Motor still on, NET 1 off
Manual Mode, GSM timeout at boot	[1,1,1,0,0,0,1,0,0,0]	Only phase + manual LEDs
Yield test running	[1,1,1,1,0,1,0,1,1,0]	Same as normal running — app banner distinguishes
OTA upgrade	[0,0,0,0,0,0,0,0,0,0]	All off during firmware flash

10. App Implementation Notes

10.1 MQTT Subscriptions Required

Topic	Purpose	QoS	Action
gw/{id}/leds	LED sequence updates	1	Update all 10 LEDs atomically
gw/{id}/telemetry	LCD text mirror + dashboard data	1	Update LCD panel + data cards
gw/{id}/status	Motor on/off events, mode changes	1	Trigger LED re-render + toasts
gw/{id}/health	Boot state, firmware version, GSM/LoRa diagnostics	0	Update connectivity indicators
gw/{id}/error	Fault alerts	1	Show fault banner, update TRIP LED
gw/{id}/response	Command ACK/NACK	1	Resolve pending button action

10.2 LCD Mirror Rules

- The app renders exactly the text published by the firmware in telemetry. It does not compute or infer LCD text independently.
- The app must support all 16 characters of Row 0 and Row 1. No truncation.
- Font: monospace (Courier or equivalent) to match physical 16×2 character display.
- Background: dark green (#33691E). Text: light green (#A5D6A7). Matches physical LCD appearance.
- Rotation timing: the firmware controls rotation. The app simply renders what it receives — it does not run its own 4-second or 3-second timer.

10.3 Button Debounce & Command Timeout

- After any button tap, disable that button for 5 seconds while waiting for response ACK.
- If no ACK received within 5 seconds, re-enable button and show toast: "Command not confirmed — device may be busy."
- Never send duplicate commands. One tap = one MQTT publish.
- START command: always show a confirmation dialog if motor has been off for less than 2 minutes (dry run protection advisory).

10.4 Dependency on Cloud Intelligence Module

- The app does not make pump start/stop decisions. It sends the user intent to the cloud.
- The cloud validates all start conditions (voltage, SDL, imbalance, connectivity) before forwarding PUMP_START to the device.
- If cloud rejects a start command, the response topic carries result: rejected with a reason message. The app displays this reason to the user.

- Yield test scheduling, duty cycle updates, and borewell science results are pushed to the app via separate dashboard API — not via MQTT directly.